

JECRC UNIVERSITY JAIPUR
Session: 2025-26
Subject: Mathematics-I (DMA024A)
Course: B. Tech - I Semester
Tutorial Sheet- 1

UNIT-1

M.M: 64

CO1: To apply differential and integral calculus to notion of curvature and to improper integrals. Apart from some other applications they will have basic understanding of Beta and Gamma function.

PART – A

(5x2=10)

Q.1 Write the formula of radius of curvature in parametric form.

Q.2 If C_x and C_y are the chords of curvature parallel the axis at any point of the curve $= ae^{\frac{x}{a}}$. Show that $\frac{1}{C_x^2} + \frac{1}{C_y^2} = \frac{1}{2aC_x}$.

Q.3 Find the radius of curvature of the Curve $S = a \log \left\{ \cot \left(\frac{\pi}{4} - \frac{\psi}{2} \right) \right\}$.

Q.4 Write the formula of radius of curvature in Polar form.

Q.5 Define the Evolutes and Involute.

PART – B

(3x7=21)

Q.1 Find the radius of curvature of the curve $x^2y = a(x^2 + y^2)$ at $(-2a, 2a)$.

Q.2 Prove that the radius of curvature of the cycloid $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$ at any point θ is $4a \cos \frac{\theta}{2}$.

Q.3 Find the center and circle of curvature of the curve $y = x^3 - 6x^2 + 3x + 1$ at the point $(1, -1)$.

PART – C

(3X11=33)

Q.1 Find the radius of curvature of the curve $x^3 + y^3 = 3axy$ at the point $(3a/2, 3a/2)$.

Q.2 show that the evolutes of the rectangular hyperbola $xy = c^2$ is the curve

$$(x+y)^{\frac{2}{3}} - (x-y)^{\frac{2}{3}} = (4c)^{\frac{2}{3}}.$$

Q.3 Prove that the radius of curvature at (x, y) on the curve $y = \frac{ax}{a+x}$ is given by

$$\left(\frac{2\rho}{a} \right)^{\frac{2}{3}} = \left(\frac{x}{y} \right)^2 + \left(\frac{y}{x} \right)^2.$$

Answer:- Part – A

3. $a \sec\phi$

Part – B

1. $-2a$

$$3.(x+36)^2 + (y + \frac{43}{6})^2 = \left(\frac{37}{36} \right)^2$$

Part – C

1. $\frac{3\sqrt{2}a}{16}$.